# Notes: Hémous, Olsen, Zanella, and Dechezleprêtre (JPE, 2025)

**Induced Automation Innovation: Evidence from Firm-Level Patent Data**

## Contents

# 1 Big picture

- **Question.** Do higher wages induce more labor-saving innovation, and if so by how much?
- **Setting.** The paper studies *automation innovation in machinery*—robots, CNC, programmable systems, automated handling and related equipment—using patent data matched to firms and country-level labor-cost data.
- **Core challenge 1: measurement.** One first needs a credible way to classify which patents are truly automation patents rather than just general machinery patents.
- **Core challenge 2: identification.** The labor costs relevant for automation innovators are not simply the wages paid by the innovating firms themselves, but the wages paid by their *downstream customers* across many countries. Those wages are endogenous and difficult to observe directly.
- **Main contribution 1.** The paper develops a transparent patent classification system for automation in machinery. It classifies technology categories rather than individual patents, using patent text at the EPO and C/IPC technology codes that can then be applied to the near-universe of patents in PATSTAT.
- **Main validation result.** Sectors that use more automation-intensive machinery experience a relative decline in routine cognitive and routine manual tasks, a rise in the skill ratio, and a fall in the labor share. So the patent classification lines up with the labor-market effects one would expect from automation.
- **Main contribution 2.** The paper estimates how firms' automation innovation responds to changes in the labor costs faced by downstream firms, using a firm-level shift-share design based on innovators' international patent exposure.
- **Main empirical result.** A 1% increase in downstream *low-skill* wages raises automation innovation by roughly 2–5%, while a 1% increase in *high-skill* wages lowers automation innovation by a similar magnitude.
- **Key comparative result.** The same wage shocks do *not* systematically affect nonautomation machinery innovation. So wages shift the *direction* of innovation toward automation, not just the overall amount of innovation.
- **Policy evidence.** Two policy-based exercises back up the main result: higher minimum wages increase automation innovation, and the German Hartz labor-market reforms reduced automation innovation by non-German firms more exposed to Germany.
- **Why the paper matters.** The paper gives direct micro evidence for induced automation innovation, a mechanism central to directed technical change, labor-market policy, and long-run growth.

# 2 Data and measurement (key objects)

## 2.1 Observed patent and text data

- **Patent sources.** The paper combines two patent datasets:

- the **EPO full-text database** (2018 release), which contains patent text in English for European patent applications;
- **PATSTAT** (autumn 2018), which contains bibliographic information on close to the universe of global patents, including dates, applicants, inventors, patent-family links, citations, and C/IPC technology codes.

- **Technology scope.** The paper focuses on *machinery patents*, which make up a large but slowly declining share of all patents. Machinery includes machine tools, handling equipment, textile and paper machines, and other special machines, plus selected combinations with control/computing codes.
- **Why patent families.** A patent family is treated as the relevant innovation unit. The date of the first application is the innovation year, and biadic patent families (applications in at least two patent offices) are used to focus on more important inventions and improve comparability across countries.

## 2.2 Observed country-level labor-market and macro data

- **Countries.** The paper uses data for 41 countries in the baseline wage exercise.
- **Main source.** Country-level labor-cost data come mainly from the 2013 release of the World Input-Output Tables.
- **Skill groups.** Low-skill workers are defined as having no high-school diploma (or equivalent), while high-skill workers have at least a college degree.
- **Wages.** The paper uses hourly labor costs in manufacturing, deflated by the local producer price index for manufacturing and converted into 1995 U.S. dollars.
- **Other macro controls.** Labor productivity in manufacturing, GDP per capita, GDP gap, and measures of manufacturing size are added to separate wage effects from general business-cycle, technology, and demand shocks.

## 2.3 Observed firm-level panel

- **Firm identification.** Firms are identified through Orbis Intellectual Property matched to PATSTAT.
- **Dependent variable.** The main left-hand-side variable is the annual *count of automation patent families* filed by a firm.
- **Market exposure weights.** For each firm, the country weights are computed from the geographic distribution of its machinery patent filings during the presample period 1971–94.
- **Why patent weights proxy for market exposure.** Firms patent in countries where they expect commercial value, so the international pattern of pre-sample patenting serves as a proxy for where their downstream market is.
- **GDP scaling.** Raw patent shares are reweighted by country size using $\mathrm{GDP}^{0.35}$, so larger markets receive more weight but not one-for-one.

- **Regression sample.** The baseline sample contains firms that applied for at least one biadic automation patent in 1997–2011 and had pre-sample international patenting. This yields 3,236 firms and captures 53.5% of global automation innovations in machinery within the relevant patent universe.

### 2.4 Unobservables and timing

- **Key unobservable.** The paper cannot observe the actual expected future labor costs faced by innovators' customers. It therefore proxies them with current country-level labor-cost shocks, combined with fixed patent-based country weights.
- **Innovation timing.** Right-hand-side variables are lagged by two years in the baseline to reflect the lag from R&D incentives to patent applications.
- **Knowledge dynamics.** Firms' own innovation stocks and external knowledge spillovers enter the regressions because innovation is path dependent and because knowledge spillovers may amplify the response of innovation over time.

### 2.5 Empirical applications

- **Application 1.** Validate the automation patent classification by linking patent use to sectoral changes in tasks, skill composition, and labor share in U.S. sectors.
- **Application 2.** Estimate the elasticity of automation innovation with respect to low-skill and high-skill wages faced by downstream firms worldwide.
- **Application 3.** Use minimum-wage changes and the German Hartz reforms as identifiable labor-market shocks.
- **Application 4.** Simulate how a persistent change in the skill premium affects the share of automation innovation and implied labor-market outcomes once knowledge spillovers are allowed to propagate.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Classifying automation technology categories

For each machinery technology category $t$, the paper computes the prevalence of automation keywords in EPO patent text:

$$s(t) = \frac{\sum_{p \in \Omega^{EPO}} \mathbf{1}\{t \in MT_p\}\, k^{any}(T_p)}{\sum_{p \in \Omega^{EPO}} \mathbf{1}\{t \in MT_p\}}.$$

- $MT_p$ is the set of machinery technology categories associated with patent $p$.
- $k^{any}(T_p) = 1$ if the patent text contains at least one automation keyword and 0 otherwise.
- **Interpretation.** This is the share of patents in technology category $t$ whose text looks "automation-like."

The paper defines:

$$t \in T^{90} \quad \text{if} \quad s(t) > 0.39, \qquad t \in T^{95} \quad \text{if} \quad s(t) > 0.48.$$

- These thresholds correspond to the 90th and 95th percentiles of the distribution of keyword prevalence across six-digit machinery technology codes.
- **Interpretation.** `auto95` is the stricter automation classification and is the paper's baseline measure; `auto90` is broader.

### 3.2 Defining automation patents

A patent family $p$ is classified as an automation patent if at least one of its machinery technology categories belongs to the automation set:

$$p \text{ is auto95 if } \exists t_p \in MT_p \text{ such that } t_p \in T^{95}.$$

- **Interpretation.** The paper first classifies *technology categories*, then maps those categories back to the global universe of patent families.
- **Advantage.** This makes the measure transparent and portable: anyone with C/IPC codes can apply it, without needing patent text for every country.

### 3.3 Sector-level validation regression

To validate the automation measure, the paper runs sector-level regressions of the form

$$\Delta T_{jk} = \beta_0 + \beta_C \, \Delta C_j + \beta_{aut} \, aut_j + \varepsilon_j. \tag{1}$$

- $\Delta T_{jk}$ is the change in task input of type $k$ in sector $j$.
- $\Delta C_j$ is Autor, Levy, and Murnane's measure of computerization.
- $aut_j$ is the paper's patent-based measure of automation exposure in the sector *of use*.
- **Interpretation.** This asks whether sectors using more automation-intensive machinery see the labor-market changes that the automation literature predicts.

### 3.4 Firm-level downstream wage measure

The average labor cost faced by firm $i$'s downstream customers is built as a shift-share weighted average of country-level labor costs:

$$w^J_{i,t} = \sum_c k_{i,c} w^J_{c,t}, \qquad J \in \{L, H\}. \tag{2}$$

- $w^L_{c,t}$ is the low-skill wage in country $c$ at time $t$.
- $w^H_{c,t}$ is the high-skill wage in country $c$ at time $t$.
- $k_{i,c}$ is firm $i$'s fixed country weight.
- **Interpretation.** A firm's relevant wage exposure is not its home wage alone, but the weighted average wage paid by its potential customers around the world.

### 3.5 Constructing patent-based country weights

Let $\tilde{k}_{i,c}$ be the fraction of firm $i$'s machinery patents protected in country $c$ during the presample period. The final country weight is

$$k_{i,c} = \frac{\tilde{k}_{i,c}\, GDP_{0,c}^{0.35}}{\sum_{c'} \tilde{k}_{i,c'}\, GDP_{0,c'}^{0.35}}. \tag{3}$$

- $GDP_{0,c}$ is the average GDP of country $c$ at the end of the presample period.
- **Interpretation.** Patenting reveals market presence, and GDP scaling adjusts for the fact that larger markets matter more, but less than proportionally.

### 3.6 Baseline Poisson patent regression

The main empirical specification is

$$\begin{aligned} \mathbb{E}(PAT_{i,t}^{Aut}) = \exp\Big(&\beta_w^L \ln w_{i,t-2}^L + \beta_w^H \ln w_{i,t-2}^H + \beta_X X_{i,t-2} \\ &+ \beta_K^a \ln K_{i,t-2}^{Aut} + \beta_K^o \ln K_{i,t-2}^{other} \\ &+ \beta_S^a \ln SPILL_{i,t-2}^{Aut} + \beta_S^o \ln SPILL_{i,t-2}^{other} + d_i + d_{j,t} (+d_{c,t})\Big). \end{aligned} \tag{4}$$

- $PAT_{i,t}^{Aut}$ is the count of automation patent families by firm $i$ in year $t$.
- $X_{i,t-2}$ includes macro controls such as labor productivity, GDP per capita, GDP gap, and in some specifications manufacturing-size controls.
- $K_{i,t-2}^{Aut}$ and $K_{i,t-2}^{other}$ are firm knowledge stocks in automation and other technologies.
- $SPILL_{i,t-2}^{Aut}$ and $SPILL_{i,t-2}^{other}$ are external knowledge spillovers.
- $d_i$ are firm fixed effects, $d_{j,t}$ are industry-year fixed effects, and $d_{c,t}$ are home-country-year fixed effects in the stricter specifications.
- **Interpretation.** The coefficient $\beta_w^L$ is the elasticity of automation innovation with respect to downstream low-skill wages; $\beta_w^H$ is the elasticity with respect to downstream high-skill wages.

### 3.7 Foreign-wage decomposition

The paper also decomposes total wage exposure into home and foreign components and uses *normalized foreign wages*. This keeps the coefficient interpretable as an elasticity with respect to total downstream wage exposure while leveraging only foreign variation once home-country-year fixed effects are included.

- **Interpretation.** This is crucial for identification because it strips out domestic shocks that could jointly affect wages and innovation in the innovator's own country.

### 3.8 Placebo regression for nonautomation innovation

The paper reruns equation (4) with dependent variables such as $pauto90$ or $pauto95$, that is, machinery innovations excluding automation patents.

- **Interpretation.** If the coefficients on low-skill wages disappear for placebo machinery patents, then the wage effect is about *directional technological change* toward automation rather than general innovative intensity.

### 3.9 Minimum-wage specification

The paper replaces low-skill wages with a shift-share measure of downstream minimum wages for the subset of countries where those data are available and otherwise keeps the baseline Poisson structure.

- **Interpretation.** This is a policy-based version of the main wage result.

### 3.10 Hartz event-study specification

To study Germany's Hartz reforms, the paper estimates

$$\mathbb{E}(PAT_{i,t}^{Aut}) = \exp\left(\beta_{DE,t}\, d_t\, k_{i,DE} + \beta_K^a \ln K_{i,t-2}^{Aut} + \beta_K^o \ln K_{i,t-2}^{other} + d_i + d_{j,t} + d_{c,t}\right).$$

- $k_{i,DE}$ is firm $i$'s fixed German exposure weight.
- $d_t$ is a full set of year dummies (with 2005 omitted in the event study).
- **Interpretation.** The coefficients trace whether firms more exposed to Germany changed automation innovation differentially around the Hartz labor-market reforms.

### 3.11 Triple-difference Hartz specification

The paper then compares automation and nonautomation machinery innovations directly:

$$\mathbb{E}(PAT_{i,t}^{k}) = \exp\left(\beta_{DE,t}\, d_t k_{i,DE} + \beta_{DE,t}^{aut}\, d_t k_{i,DE} \mathbf{1}_{k=aut} + \cdots + d_{k,i} + d_{k,j,t} + d_{k,c,t}\right). \tag{5}$$

- $k \in \{auto95, pauto95\}$ (or $pauto90$ in an alternative version).
- The coefficient $\beta_{DE,t}^{aut}$ measures whether German exposure changes the *relative propensity* to innovate in automation rather than in other machinery technologies.
- **Interpretation.** This is the cleanest event-study evidence that the Hartz reforms shifted the *direction* of innovation.

### 3.12 Simulation of a permanent skill-premium shock

Using estimated Poisson equations for automation and nonautomation machinery patents, the paper simulates the response to a permanent 10% fall in the skill premium.

- First it computes the direct effect of wages on patent counts.

- Then it updates firms' own innovation stocks.
- Finally it recomputes knowledge spillovers over time.
- **Interpretation.** This translates the estimated innovation elasticities into medium-run changes in the *share* of automation innovation and then into labor-market outcomes using the earlier sector-level mapping.

# 4 Identification and instruments (what makes the estimates credible)

## 4.1 What fails in a naive interpretation

- Looking at raw correlations between domestic wages and innovation would be misleading because domestic tax, demand, and technology shocks can affect both wages and innovation.
- Using actual customer wages directly would also raise reverse-causality problems.
- Country composition matters: firms sell to different sets of countries, so any empirical design must distinguish wage shifts from firm exposure shares.

## 4.2 What identifies the model in the new approach

- The paper includes firm fixed effects, industry-year fixed effects, and in the core specifications home-country-year fixed effects.
- After these controls, the identifying variation comes from *foreign* country-level shocks interacting with fixed pre-sample firm exposure weights.
- Because the regressions also control for high-skill wages, the coefficient on low-skill wages is identified from foreign shocks with *asymmetric* effects on low- and high-skill labor costs.

## 4.3 Why the shift-share design is plausible

- The paper interprets identification using the Borusyak-Hull-Jaravel shift-share framework: conditional on controls and weights, wage shocks must be as-good-as-randomly assigned.
- The exposure weights are not overly concentrated. The Herfindahl index of foreign country weights is about 0.09 at the country level and about 0.006 at the country-year level.
- Pairwise weight correlations are low, which supports the "many weakly correlated shocks" requirement for shift-share identification.

## 4.4 How the paper addresses alternative channels

- **Demand shocks.** The paper controls for GDP gap, GDP per capita, labor productivity, manufacturing size, and low-skill-weighted manufacturing size. The wage coefficients remain stable.

- **Technology shocks and reverse causality.** The paper controls for recent automation and other innovation, firms' own knowledge stocks, and spillovers. The low-skill wage coefficient remains stable, and if anything reverse causality would bias it downward.
- **Skill-neutral innovation.** This is handled by comparing automation patents with nonautomation machinery patents generated by the same firms for the same broad technological areas.

### 4.5 Why the placebo tests are informative

- The low-skill and high-skill wage coefficients are strong and precisely estimated for `auto95` patents.
- They are much smaller and mostly insignificant for `pauto90`, refined `pauto90`, and `pauto95` patents.
- This means the main regressions are picking up induced *automation* innovation, not general machinery innovation.

### 4.6 Why the policy shocks strengthen credibility

- Minimum wages are a direct policy lever affecting low-skill labor costs.
- Germany's Hartz reforms are a large, salient labor-supply and labor-cost shock that is unlikely to affect foreign firms' innovation through other channels once German firms themselves are excluded.
- Finding results of similar sign and magnitude in these policy exercises makes the baseline interpretation much more persuasive.

### 4.7 Relation to the literature

- The paper contributes to the literature on **induced innovation** by showing that labor costs alter the direction of innovation toward automation.
- It contributes to the literature on **automation measurement** by constructing a new patent-based automation classification for machinery.
- It also complements work on **technology adoption** by focusing on innovation rather than adoption, which is more relevant for long-run growth and knowledge-spillover dynamics.

## 5 Tables and figures

### 5.1 Table 1: Automation keywords

**Read:**

- The classification is not an opaque machine-learning black box. It is built from economically interpretable keyword families such as *robot*, *numerical control*, *computer-aided design/manufacturing*, *flexible manufacturing*, *programmable logic control*, *3D printing*, and terms related to *automation* and *labor*.

- The paper deliberately ties most keywords to earlier automation measurement work, then refines them with context words to avoid false positives.
- The point of the table is transparency: readers can see exactly what counts as "automation."

## 5.2 Figure 1: Prevalence of automation keywords across machinery technology codes

**Read:**

- The distribution of automation prevalence across machinery technology categories is highly skewed.
- Most codes have low automation intensity, but a small upper tail looks strongly automation-related.
- This justifies the percentile-threshold approach used to define `auto90` and `auto95`.

## 5.3 Figure 2: Trends in automation patents

**Read:**

- Machinery patents account for a sizable share of all patents, around 18.6% on average, though that share drifts down slowly over time.
- Within machinery, the automation share falls slightly from the mid-1980s to the mid-1990s and then rises sharply afterward. For `auto95`, the worldwide share goes from about 9.4% in 1985 to 7.5% in 1994 and then up to 19.1% by 2015.
- Japan leads early in the sample, while Germany becomes especially important from the 2000s onward.
- So automation is not just a fixed patent category; it becomes increasingly important over time inside machinery innovation.

## 5.4 Table 2 and Table 3: Sector validation

**Read:**

- Table 2 shows large cross-sector variation in the share of automation patents used by sectors.
- Table 3 is the core validation exercise. A 1 percentage point increase in a sector's automation-patent share is associated with about a 1.4-centile decline in routine cognitive tasks and a 1.3-centile decline in routine manual tasks per decade.
- The same increase is associated with a sizable rise in the skill ratio and about a 1.3 percentage point decline in the labor share in manufacturing sectors.

- These are exactly the patterns one wants from a measure of automation: fewer routine tasks, more skill intensity, and lower labor share.

## 5.5 Figure 3 and Table 4: Wage variation and firm exposure

**Read:**

- Figure 3 shows there is meaningful cross-country and time variation in low-skill wages and in the inverse skill premium. For example, the United States sees a relative decline in the inverse skill premium while Italy follows a more nonmonotonic path.
- Table 4 shows firms are internationally exposed but not uniformly globalized. The average largest-country weight is 0.47 and the second-largest is 0.17.
- The top weighted countries are the United States, Germany, Japan, and France.
- The weights are diversified enough for a shift-share design: foreign-weight concentration is modest and pairwise correlations are low.

## 5.6 Table 5: Baseline effect of wages on automation innovation

**Read:**

- This is the paper's main table.
- A 1% increase in downstream low-skill wages raises automation patenting by roughly 2.7–3.6% in the baseline specifications and by about 4.2–5.3% when the design focuses on normalized foreign wages.
- A 1% increase in downstream high-skill wages lowers automation innovation by roughly similar magnitudes.
- The results survive country-year fixed effects, so they are not driven by domestic co-movement between wages and innovation.
- Knowledge spillovers matter a lot: spillover variables enter positively and significantly, while own automation stock enters negatively, suggesting diminishing private returns but strong externalities.

## 5.7 Table 6 and Table A10 logic: placebo machinery innovations

**Read:**

- When the dependent variable is nonautomation machinery patenting, the wage coefficients become small and mostly insignificant.
- The same firms, facing the same international environment, do *not* increase generic machinery innovation in response to low-skill wage shocks.
- That is the paper's strongest evidence that wages change the *direction* of innovation toward automation.

### 5.8 Table 7 and Table 8: alternative channels and falsification

**Read:**

- Table 7 adds controls for manufacturing size, low-skill-weighted manufacturing demand, recent innovation, and technology shocks. The main coefficients barely move.
- This weakens stories based on foreign demand shifts or omitted technology shocks.
- Table 8 implements permutation-based falsification and inference exercises tailored to the shift-share setting. The main coefficients remain significant, especially in the foreign-wage specifications.
- So the result is not a fragile artifact of correlated residuals or concentrated exposure weights.

### 5.9 Figure 4: simulated effect of a permanent fall in the skill premium

**Read:**

- A permanent global 10% decrease in the skill premium raises the average share of automation innovation in machinery by 4.3 percentage points between 1997 and 2011.
- Of that, 2.3 percentage points come from dynamic amplification through knowledge spillovers beyond the direct effect.
- Mapping this through the sector-level estimates implies large labor-market responses: routine cognitive tasks fall by about 6.1 centiles, routine manual tasks by 5.5 centiles, and the manufacturing labor share by 5.6 percentage points per decade.
- This figure is the bridge from micro patent elasticities to medium-run macro labor-market effects.

### 5.10 Table 9, Figure 5, and Table 10: policy shocks

**Read:**

- Table 9 shows that higher downstream minimum wages also increase automation innovation, with coefficients broadly similar to those on low-skill wages in the baseline regressions, though less precise in the foreign-only specifications.
- Figure 5 studies the German Hartz reforms. Firms more exposed to Germany show a reversal in automation innovation after the reforms, exactly when Germany's low-skill labor costs stopped rising relative to the rest of the world.
- The 2010 event-study coefficient implies that a firm with German exposure of 0.1 had about a 27.1% smaller increase in automation innovations between 2005 and 2010 than a firm with no German exposure.
- Table 10 shows this effect is specific to automation innovation relative to other machinery innovation: the post-2005 trend break in the triple-difference specification is about $-1.05$ to $-1.12$ and highly significant.

# 6 Short summary

- **Measurement result.** The paper builds a new patent-based measure of automation innovation in machinery that behaves exactly as one would expect in sector-level labor-market data.
- **Causal result.** Firms innovate more in automation when the low-skill labor costs of their downstream customers rise, and less when high-skill labor costs rise.
- **Direction-of-innovation result.** These wage effects are specific to automation patents, not general machinery patents.
- **Policy result.** Minimum wages push automation innovation up, while the German Hartz reforms—which lowered the effective cost of low-skill labor—pushed it down for firms exposed to Germany.
- **Broader lesson.** Labor-market policy and labor-supply shocks do not just affect employment and wages directly; they also reshape the direction of technological change.

# 7 Conclusion

## 7.1 Core conclusion (what the paper establishes)

- The paper establishes a credible firm-level link from downstream labor costs to automation innovation.
- It shows that gross shifts in low-skill versus high-skill labor costs change the composition of machinery innovation, not merely the level of innovative activity.
- It also provides a new, operational automation-patent classification that other researchers can use in future work.

## 7.2 Economic intuition

- Automation equipment producers sell to firms that are trying to economize on expensive low-skill labor and to complement scarce high-skill labor.
- When low-skill labor becomes more expensive, the market for automation expands, and innovators redirect research effort toward automation technologies.
- Because innovation builds on past knowledge and on spillovers from others' discoveries, the effect is not one-shot. A wage shock today changes the future path of innovation.

## 7.3 Takeaway

This paper is important because it turns a classic theoretical claim—that higher wages induce labor-saving innovation—into a convincing empirical result. It does so with a careful automation measure, a firm-level shift-share design, strong placebo tests, and concrete policy shocks. The paper's larger message is that labor-market institutions and technological change are jointly determined: changes in wages today help shape the machines that firms invent tomorrow.